

Lecture 2: HVS, Image Sensing and Digitization

HVS Client (Human Visual System Client) means the system or agent that receives and interprets visual information.

It can be:

- i. **Human** – The biological **Human Visual System (HVS)** (eyes + brain) processes images and understands objects, colors, motion, etc.
- ii. **Self-driving car** – Example: Tesla autonomous cars use cameras and computer vision algorithms to detect roads, pedestrians, and traffic signs. Here, the “client” is an AI-based vision system.
- iii. **Automated machine** – Industrial robots or inspection systems use machine vision to identify defects, measure position, and control operations.

So, an HVS client can be **a human or any intelligent vision-based system (AI/machine vision system)** that interprets visual data.

The **main parts of the visual system** are:

Eye – Sensor of the HVS

- Captures light from the environment
- Converts light into electrical signals

Brain – Image Processing Unit

- Processes and interprets visual signals
- Recognizes color, shape, motion, and depth
- Integrates intelligence and past experience with visual input

So, the eye collects the data, and the brain analyzes it using knowledge and experience.

Brain fills the information gap

- The brain does not receive perfect or complete visual data.
- It uses past experience, memory, and intelligence to fill missing or unclear information.
- That's why sometimes we see optical illusions or misinterpret images.

So, the brain not only processes images - it also *predicts and completes* missing information.

Important Parts of the Eye in HVS

- **Iris & Pupil**

- Control the amount of light entering the eye.
- The pupil size changes

(small in bright light, large in dim light).

- **Ciliary Body**

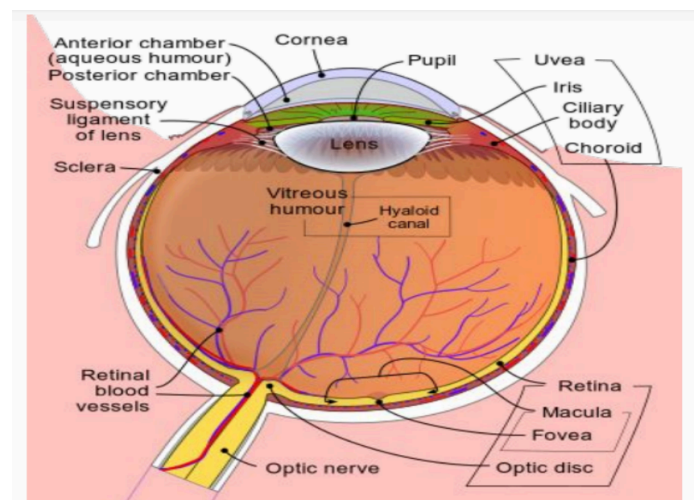
- Adjusts the focal length of the optical lens.
- Helps focus on near and distant objects (accommodation).

- **Retina**

- Acts as the image sensor.
- Contains two types of photoreceptors:
 - **Rods** – Sensitive to low light (night vision, black & white).
 - **Cones** – Responsible for color vision and fine details.

- **Fovea**

- The central focusing region of the retina.
- Contains a high concentration of cones.
- Provides the sharpest and clearest vision.



These parts work together to capture and process visual information in the Human Visual System (HVS).

Photoreceptors in the Retina

Cones

- Low in number compared to rods
- Concentrated in the central region called the **fovea**
- Highly sensitive to color
- Responsible for fine details and high resolution
- Each cone is connected to a single nerve fiber → sharp and detailed vision

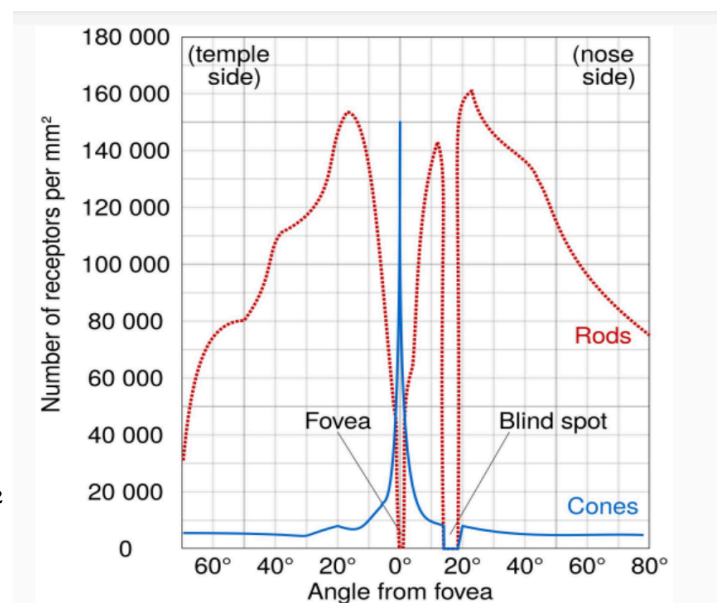
Rods

- High in number
- Distributed over most of the retina (except the fovea)
- Very sensitive to low light (night vision)
- Do not detect color (black and white vision)
- Several rods connect to one nerve fiber → lower resolution

So, **cones = color and sharp vision**, while **rods = night and low-light vision**.

Cones Distribution in the Retina

- The number of cones increases sharply toward the **fovea**.
- There is a peak density at the center of the fovea.
- There are about 150,000 cones per mm^2 in the fovea.



Rods Distribution

- Rods are more or less uniformly distributed over most of the retina.
- In the **fovea**, the number of rods becomes almost **zero**.
- This is because the central fovea contains only cones - rods and cones do not occupy the exact same spot at the very center.

So, the fovea is specialized for **sharp, color vision (cones only)**, while rods are mainly responsible for **peripheral and low-light vision**.

Blind Spot

- **Location:** Optic disc (where the optic nerve exits the eye)
- **Reason:** No rods or cones present
- **Effect:** Cannot detect light in this area

Brightness Adaptation

- **Camera vs Human Vision:**
 - Cameras capture **intensity** (objective)
 - Humans perceive **brightness** (subjective)
- **Perception:**
 - Subjective brightness is a **logarithmic function** of light intensity
- **Adaptation Range:**
 - **Photopic vision** (daylight) → high adaptation range
 - **Scotopic vision** (low light) → low adaptation range
- **Discrimination:**
 - For a given brightness level (e.g., **B_a**), the **Human Visual System (HVS)** can discriminate only a **small range of intensities** (shown as **B_a–B_b**)

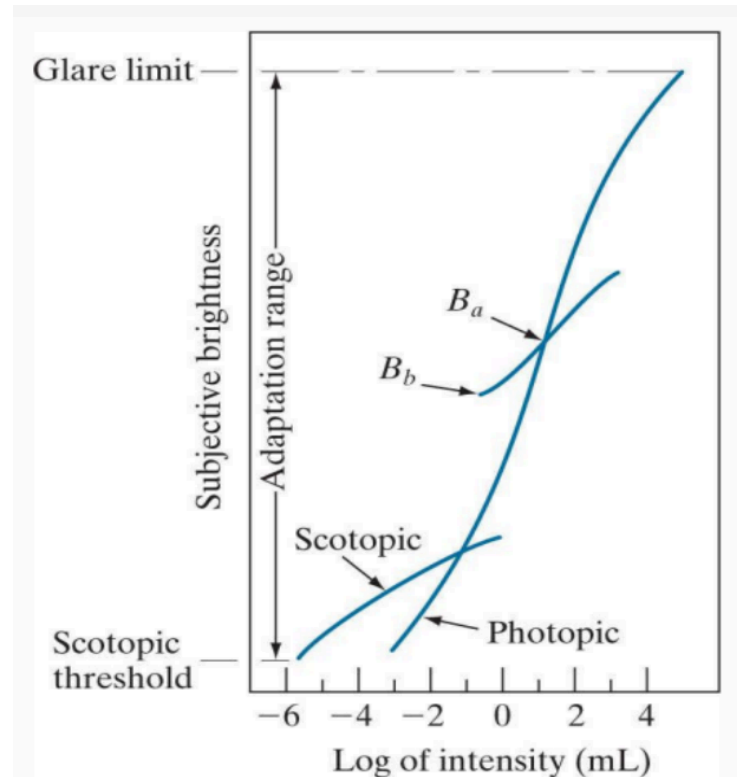
Curves

- **Scotopic Curve:**

- Represents **low-light (night) vision**, dominated by rods.
- Adaptation range is **narrow**, so small intensity changes are noticeable in dark conditions.

- **Photopic Curve:**

- Represents **daylight vision**, dominated by cones.
- The adaptation range is **wide**, so humans can handle very bright light without saturation.



Brightness Discrimination:

- **I** = background intensity
- **ΔI** = smallest intensity change detectable (**just noticeable difference, JND**)

Weber Ratio:

Weber ratio = $\Delta I / I$

- **Smaller Weber ratio** → tiny changes in intensity can be noticed → **better brightness discrimination**
- **Larger Weber ratio** → need bigger intensity changes to notice → **poorer discrimination**

Example:

- Dim background: $\Delta I = 1$, $I = 10 \rightarrow$ Weber ratio = 0.1 (good discrimination)
- Bright background: $\Delta I = 5$, $I = 100 \rightarrow$ Weber ratio = 0.05 (even better discrimination)

Weber Ratio & Brightness Discrimination

- **Weber ratio ($\Delta I/I$)** measures the smallest detectable brightness change.
- **As background intensity (I) increases**, Weber ratio decreases $\rightarrow$ **better brightness discrimination**.
- **Poorly illuminated images** have low $I \rightarrow$ Weber ratio is high $\rightarrow$ small differences are hard to see $\rightarrow$ **contrast stretching** is needed to enhance visibility.

Note: Higher intensity $\rightarrow$ finer brightness perception; low intensity $\rightarrow$ need contrast enhancement.

Brightness Discrimination & Background

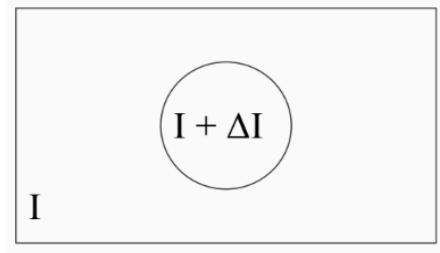
- **The Weber ratio ($\Delta I/I$)** depends on background intensity (I).
- On a **bright (white) background**: I is high $\rightarrow \Delta I/I$ is small $\rightarrow$ we can detect **small changes in brightness** easily.
- On a **dark (black) background**: I is low $\rightarrow \Delta I/I$ is large $\rightarrow$ small changes are **hard to notice** $\rightarrow$ poor brightness discrimination.

Key idea: Bright backgrounds enhance our ability to see small differences; dark backgrounds make it harder.

◆ Weber Ratio (Brightness Perception)

- Let
 - I = background intensity
 - ΔI = minimum change in intensity detectable (Just Noticeable Difference, JND)
- **Weber Ratio** is defined as:

$$\text{Weber ratio} = \Delta I / I$$



◆ Interpretation

- **Small Weber ratio**
 - very small intensity change is noticeable
 - **high brightness discrimination (good vision)**
- **Large Weber ratio**
 - larger change needed to notice difference
 - **poor brightness discrimination**

◆ Relationship with Intensity

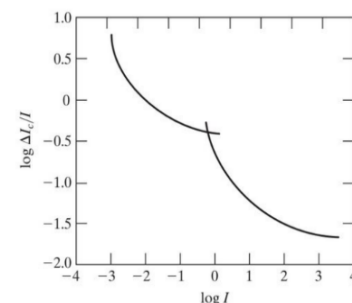
- As **background intensity (I)** increases:
 - Human eyes become **better at distinguishing changes**
 - Required relative change ($\Delta I / I$) becomes effectively **smaller in perception**

So,

- Higher intensity → better discrimination
- Lower intensity → poor discrimination

◆ Why Contrast Stretching is Needed

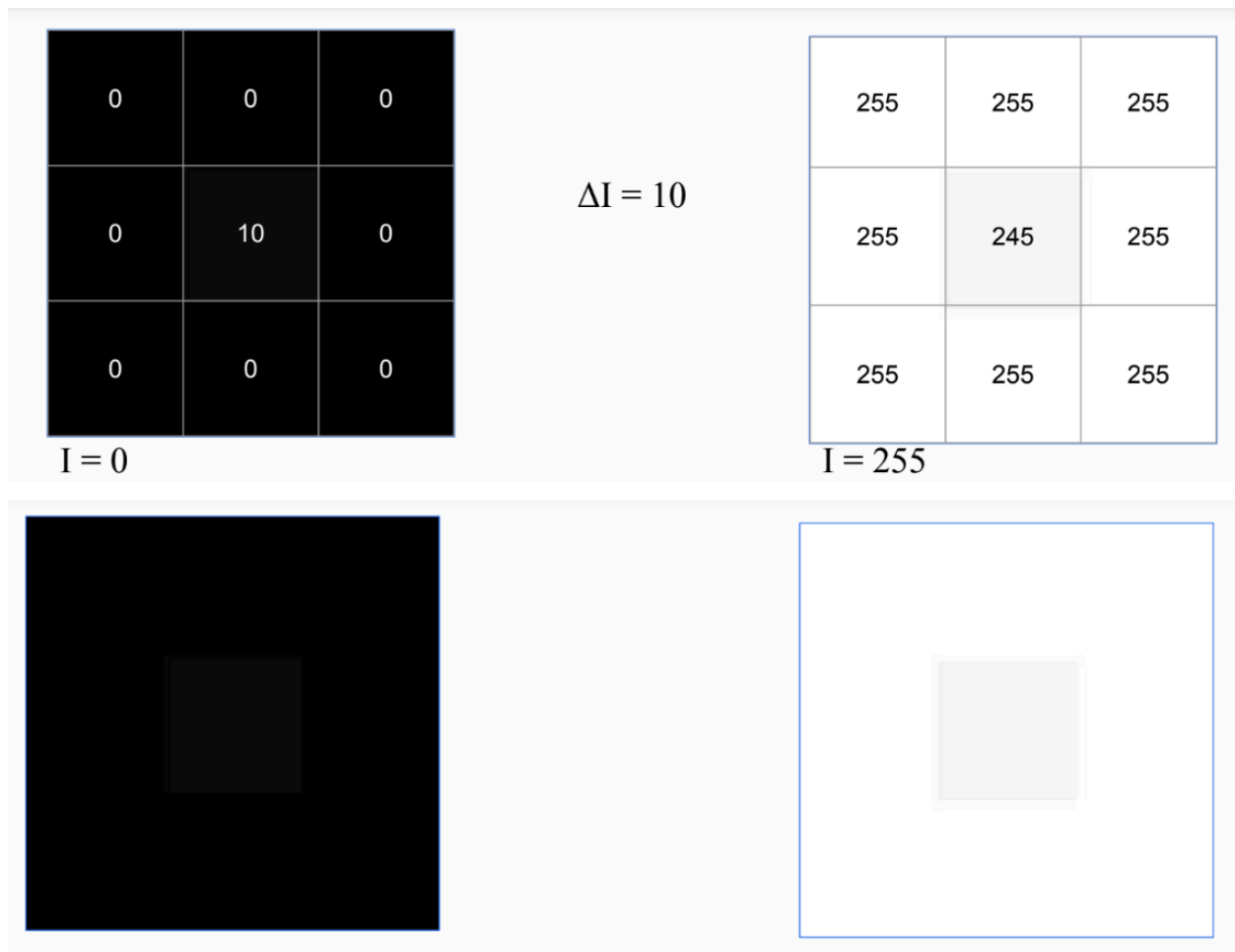
- In **poorly illuminated images (low I)**:
 - Weber ratio becomes large
 - Small differences are **not visible**



- So we apply **contrast stretching**:
 - Expands intensity range
 - Makes small differences more visible

Intuition

- In a **dark room** → hard to notice small brightness changes
- In a **well-lit room** → even tiny differences are visible



♦ Given:

- $\Delta I = 10$ (same change in intensity in both cases)

♦ Case 1: Black (dark) background

- Background intensity **I** is small

- Weber ratio:
- $\Delta I / I = 10 / \text{small} \Rightarrow \text{large value}$

Large Weber ratio $\rightarrow$ **poor discrimination**

So, we **hardly notice the change**

♦ Case 2: White (bright) background

- Background intensity **I is large**
- Weber ratio:
 $\Delta I / I = 10 / \text{large} \Rightarrow \text{small value}$

Small Weber ratio $\rightarrow$ **good discrimination**

So, we **clearly notice the change**

♦ Final Conclusion

Even though $\Delta I = 10$ is same, visibility depends on **I**:

- Dark background $\rightarrow$ change looks insignificant
- Bright background $\rightarrow$ change becomes noticeable

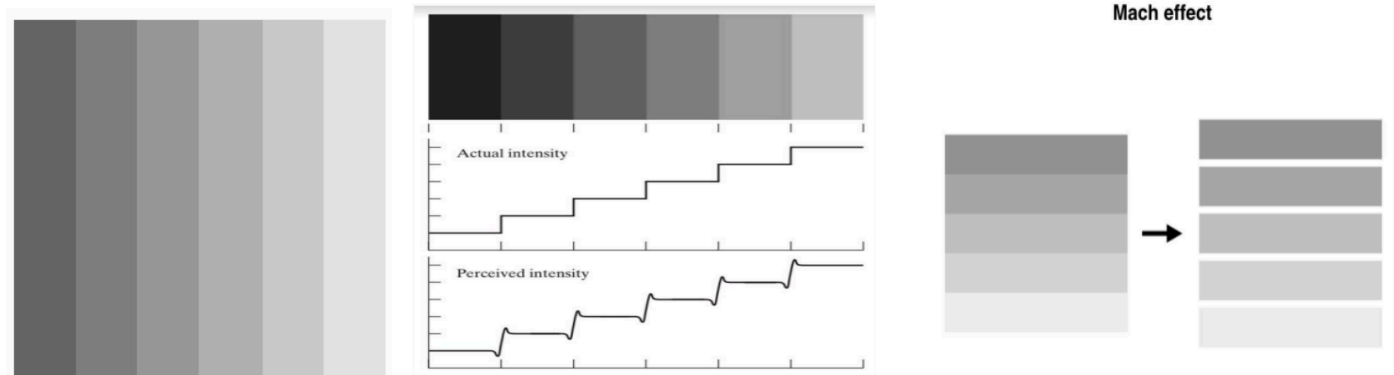
♦ Mach Bands

Mach Bands are a visual illusion where the human eye **perceives bright and dark bands near intensity boundaries**, even though they are **not actually present** in the image.

♦ What happens?

- At the boundary between a **dark and bright region**:
 - The **bright side looks brighter**
 - The **dark side looks darker**

This creates artificial “bands” at edges.



Optical Illusion Due to Perceived Brightness

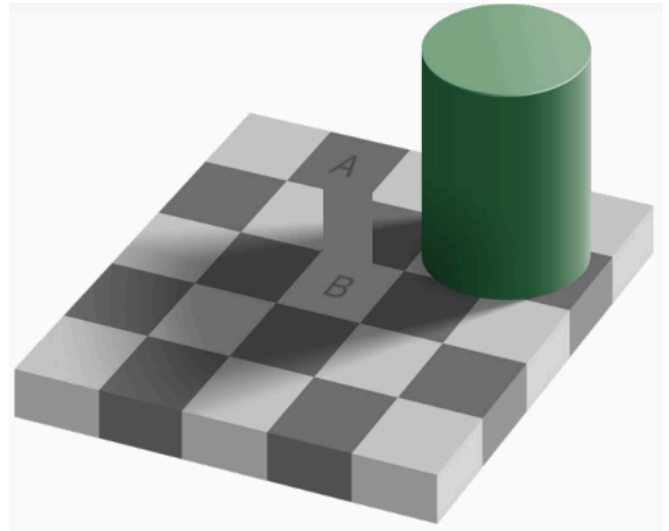
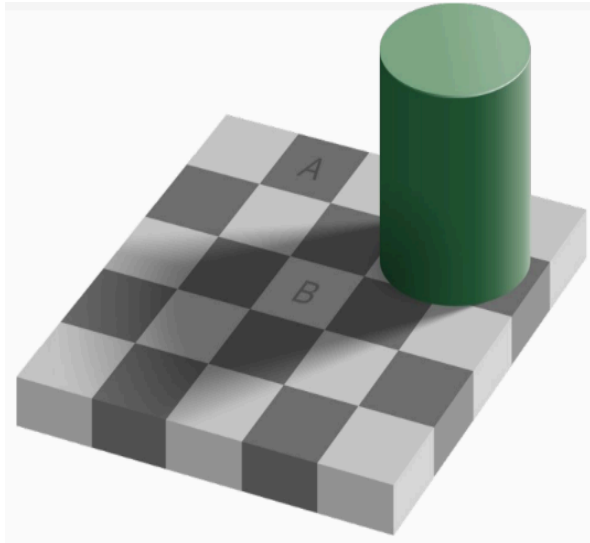


Identify the middle-box with the highest intensity



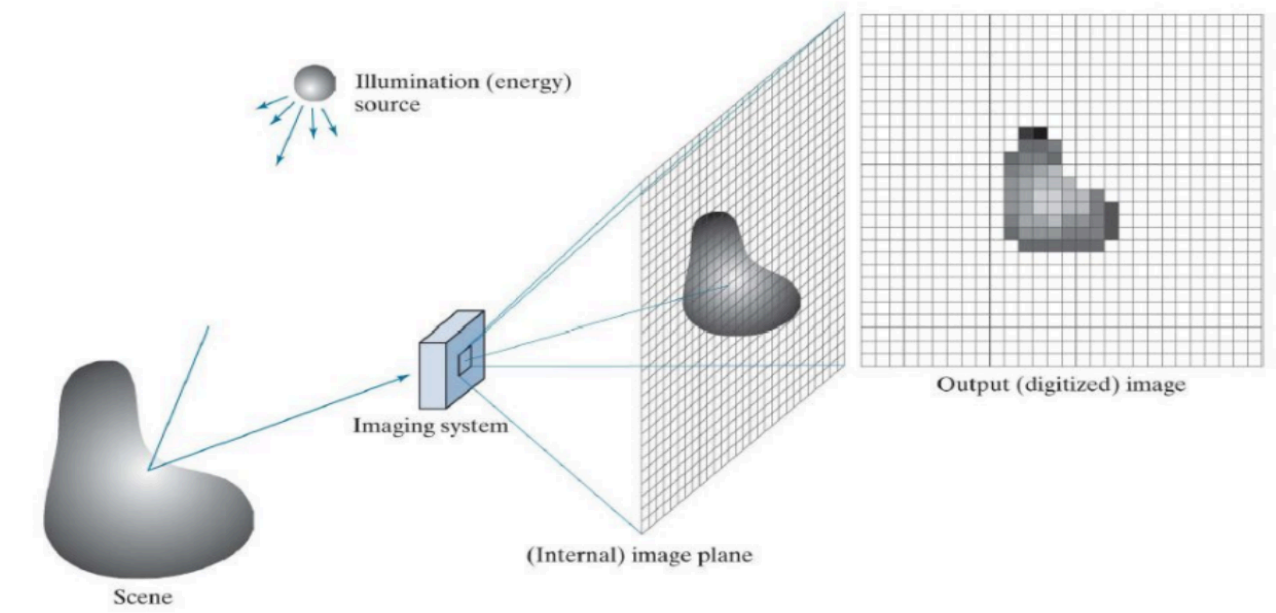
They have the same intensity!

Even if two regions have the same intensity, they may **appear different** depending on surroundings.



They have the exact same intensity!

Digital Image Acquisition



This image illustrates the fundamental process of **Digital Image Acquisition**, which is how a physical scene is converted into a digital format that a computer can process.

1. Image Formation

Everything begins with an **illumination source** (like the sun or a light bulb). This energy strikes the "Scene" (the object). The object reflects a certain amount of that light toward the **Imaging System** (the camera lens). The lens collects these light rays and projects them onto an **internal image plane**.

2. Sensing

The internal image plane is where the **CCD (Charge-Coupled Device) sensor array** lives.

- The sensor is made of a grid of many tiny photosensitive elements.
- When light hits these elements, they produce an electrical voltage proportional to the light intensity.
- At this stage, the image is still **continuous** (analog).

3. Digitization

To make the image "digital," the analog signal must undergo two specific sub-processes:

- **Sampling:** Measuring the signal at discrete spatial intervals (creating the grid of pixels).
- **Quantization:** Converting the continuous voltage levels into discrete integer values (e.g., 0 to 255 for an 8-bit image).

4. Representation

The final result is the **Output (digitized) image**. In computer memory, this is represented as a **2D matrix** of numbers, where:

- $f(x, y)$ represents the intensity (brightness) of the image at the spatial coordinates (x, y) .
- Each "square" in the grid becomes a **pixel** (picture element).

Summary of the Pipeline

Stage	Input	Output
Formation	Light & Object	Continuous Optical Image
Sensing	Optical Image	Analog Electrical Signal
Digitization	Analog Signal	Digital Bitstream
Representation	Digital Bits	2D Numerical Matrix (Pixels)

Image Sensing Pipeline

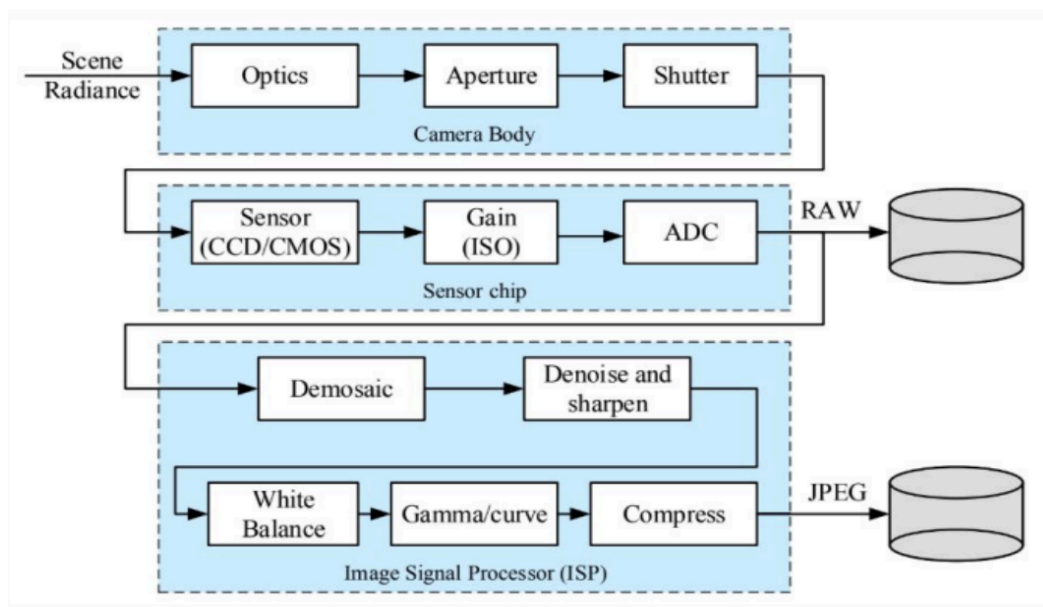
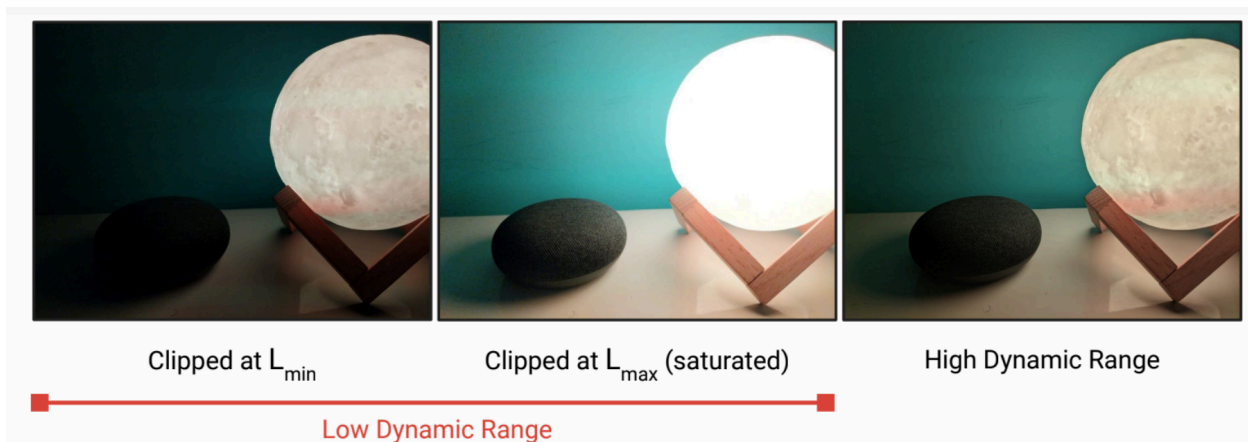


Image Formation Model

- The captured intensity $0 \leq f(x, y) < \infty$
- $f(x, y) = i(x, y) r(x, y)$
- Illumination, $0 \leq i(x, y) < \infty$
- Reflectance, $0 \leq r(x, y) \leq 1$
- $L_{\min} \leq f(x, y) \leq L_{\max}$. The ratio $L_{\max} / L_{\min}$ is called the dynamic range.
- The interval $[L_{\min}, L_{\max}]$ is called gray scale. Typical indoor values, $L_{\min} \approx 10$, $L_{\max} \approx 1000$



1. Clipped at $L_{\min}$ (Under-exposed)

In the first image, the camera sensor did not receive enough light, or the sensitivity was set too low.

- **The Result:** The dark areas of the scene (like the shadows and the object on the left) have intensity values that fall below the sensor's minimum detectable threshold ($L_{\min}$).
- **The Problem:** These areas are "clipped" to pure black. All detail in the shadows is lost because the sensor cannot distinguish between different levels of very low light.

2. Clipped at $L_{\max}$ (Over-exposed / Saturated)

In the middle image, the sensor was exposed to too much light.

- **The Result:** The bright areas (the moon lamp and the wall behind it) have intensity values exceeding the sensor's maximum capacity ($L_{\max}$).
- **The Problem:** These pixels are "saturated" and clipped to pure white. We lose the texture on the surface of the moon lamp because the sensor is "blinded" by the intensity, hitting its upper limit.

3. High Dynamic Range (HDR)

The final image represents a **High Dynamic Range** capture.

- **The Solution:** This is often achieved by combining multiple exposures (like the first two) or using a superior sensor.
- **The Benefit:** It captures a wider span of light intensities. Notice how we can see both the texture of the dark object on the left and the craters on the bright moon lamp simultaneously. It manages to keep the intensity values within the range $[L_{\min}, L_{\max}]$ without hitting the "ceiling" or the "floor."

Image Sensing - Single Sensing Element

Basic Idea

- An image sensor (like a CCD) converts **light** → **electrical signal** → **digital image**

Sensor Array

- A typical CCD camera has **4000 × 4000 sensors (pixels)**
- Each sensor:
 - Captures light from a small area
 - Produces one pixel value

How Each Sensor Works

- The electrical response of a sensor is:
Output $\propto \int$ light intensity over time

Meaning:

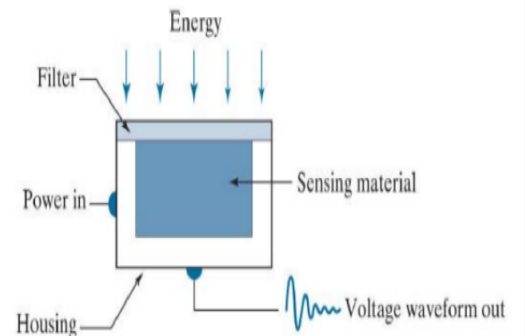
- It **integrates light over exposure time**
- More light or longer exposure $\rightarrow$ stronger signal

Why Integration Helps

- Integration over time:
 - **Averages out noise**
 - Improves signal quality

Digitization

- The analog electrical signal is:
 - Converted to digital values (ADC)
 - Stored as a **digital image (matrix of pixels)**



♦ Shutter Mechanisms

1. Rolling Shutter

- Pixels are exposed **row by row (sequentially)**

Characteristics:

- Different rows captured at **slightly different times**
- Faster and cheaper

Problem:

- **Distortion in moving objects**
 - Example:
 - Moving car looks **skewed**
 - Rotating fan looks **bent**

2. Global Shutter

- All pixels are exposed **at the same time**

Characteristics:

- Captures the entire frame **instantaneously**

Advantage:

- No motion distortion
- Accurate for fast-moving scenes

Drawback:

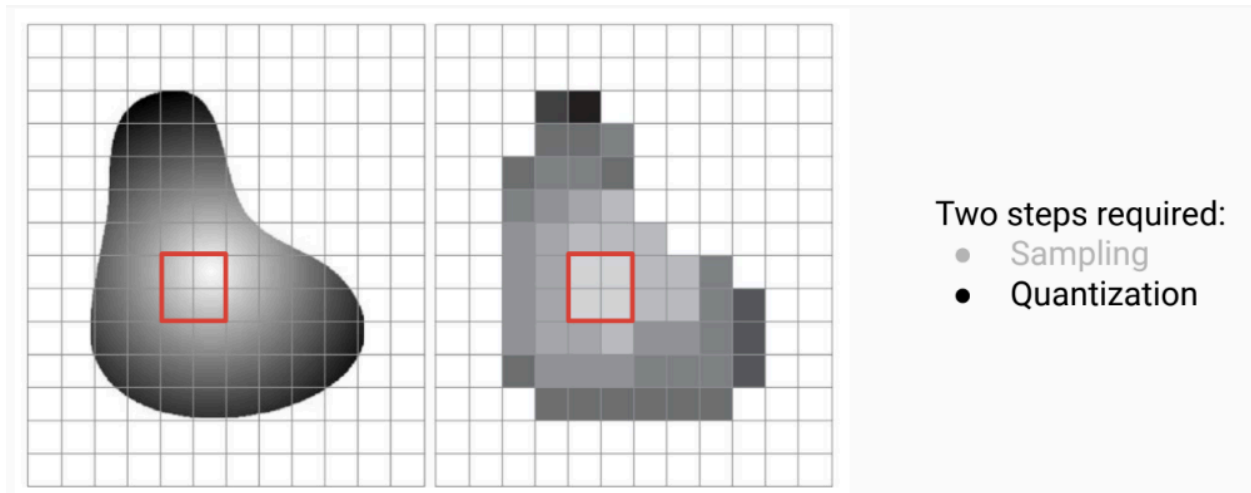
- More complex and expensive

Feature	Rolling Shutter	Global Shutter
Exposure	Row-by-row	All at once
Motion distortion	Yes	No
Cost	Low	High
Speed	High	Moderate

♦ Intuition

- **Rolling shutter** → like scanning a page line by line
- **Global shutter** → like taking a snapshot instantly

Digitalization



1. Sampling (Spatial Digitization)

Sampling is the process of dividing the continuous 2D plane into a grid of discrete points or "cells."

- **The Grid:** In the image, we can see a grid overlaid on the shape. Each square in that grid represents a **pixel** (picture element).
- **What it does:** It determines the **spatial resolution** of the image.
- **The Result:** Instead of an infinite number of points, we now have a finite MxN matrix of locations. If we sample more frequently (smaller squares), the image looks sharper and more like the original.

Think of it like cutting a pizza:

- Cut into **4 big slices** → less detail
- Cut into **100 small slices** → more detail

Same pizza, but **more pieces = better representation**

So:

- Sampling = **how many pixels (grid size)**
- More sampling → **sharper image**
- Less sampling → **blurry/blocky image**

2. Quantization (Intensity Digitization)

Once we have the grid, we need to decide what "value" to assign to each square. Real-world light intensity is continuous, but computers use a finite set of numbers.

- **The Grayscale:** Notice how the right-hand image uses blocks of specific gray shades.
- **What it does:** It maps the continuous range of brightness (amplitudes) into discrete levels (e.g., 0 to 255 for an 8-bit image).
- **The Result:** It determines the **color or gray-level depth**. If we use too few levels, we get "contouring," where the smooth gradients look like distinct steps or bands.

Imagine we are drawing a sunset.

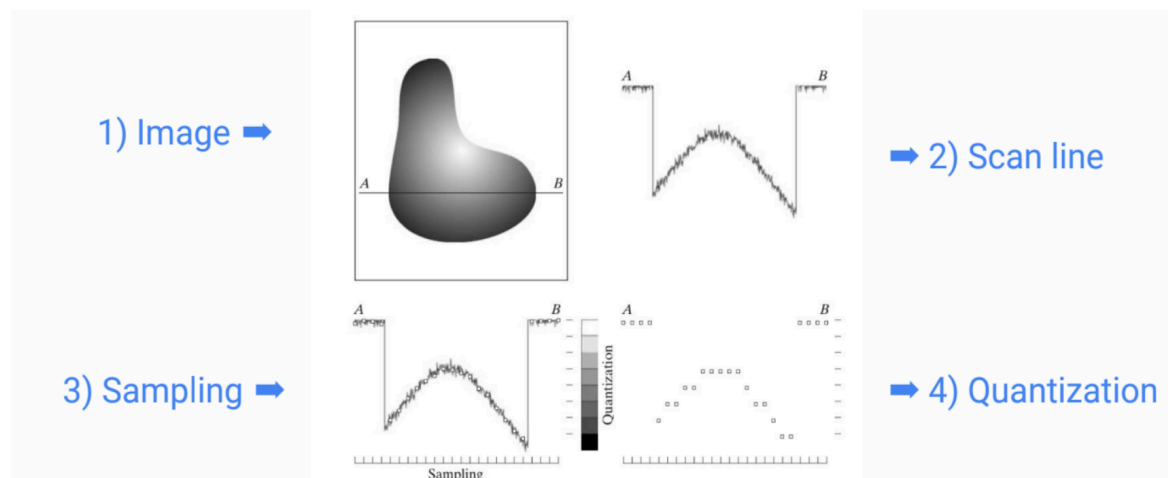
Case 1: We have only 3 colors

- Dark
- Medium
- Light

our drawing will look **rough and banded**

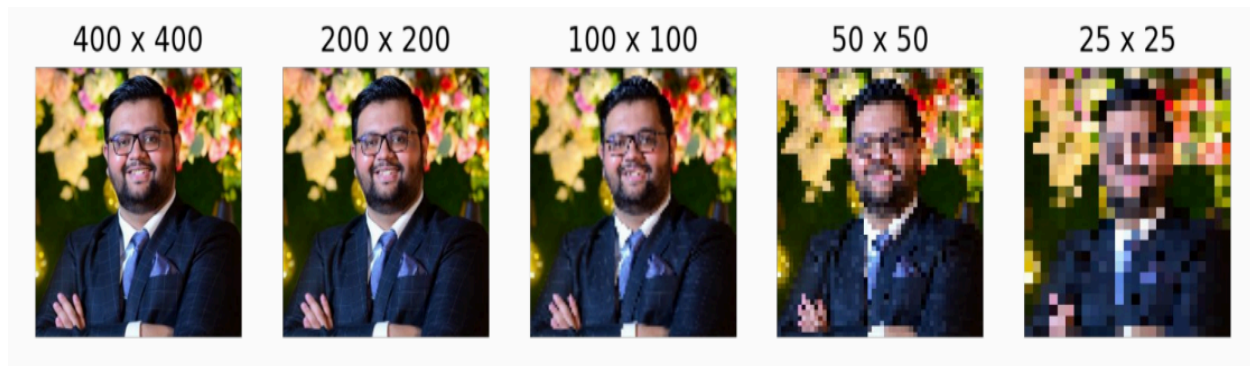
Case 2: We have 100 colors

our drawing will look **smooth and realistic**

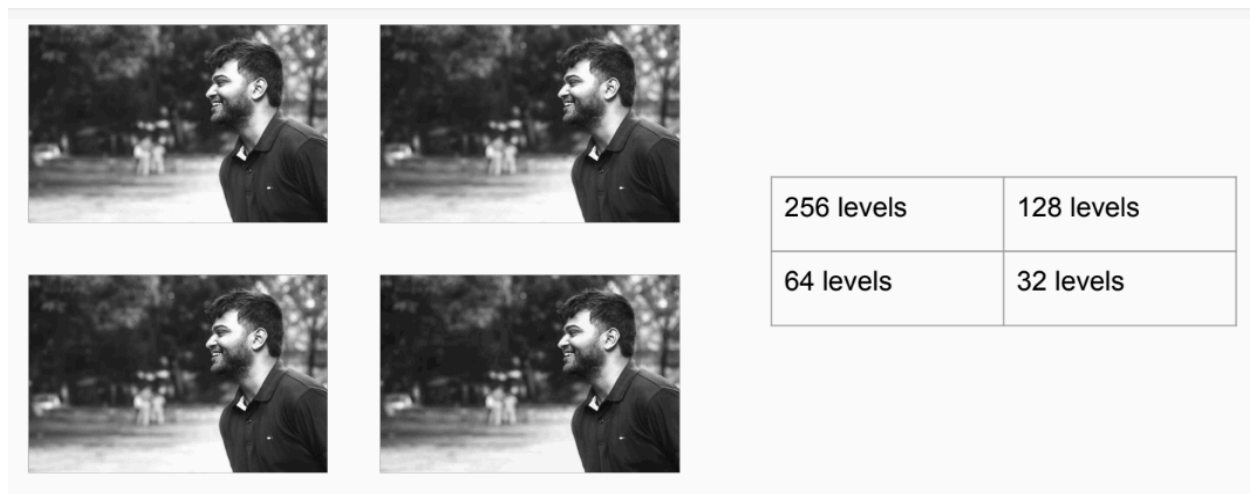


- **Sampling = Quantity of pixels (resolution)**
→ “How many squares?”
- **Quantization = Quality of color/brightness**
→ “How many shades per square?”

Effect of Sampling



Effect of Quantization



♦ False Contouring (Black-White Representation)

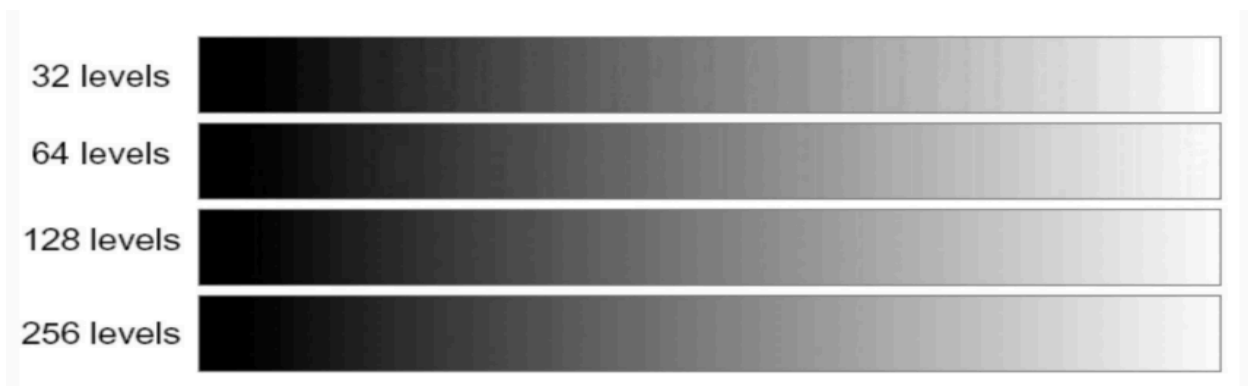
When an image is **heavily quantized** (very few intensity levels), smooth variations are forced into a **small number of colors**. For example:

- Only **black and white** (2 levels)

♦ What happens?

- All pixel intensities are mapped to:
 - **Black (0)** or
 - **White (1)**

No intermediate gray levels.



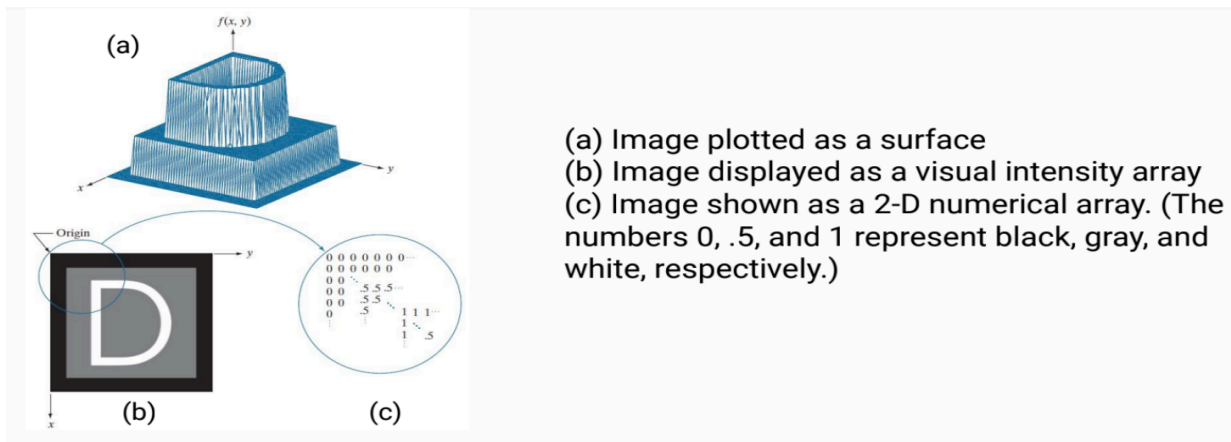
Representation

- $L_{\min} \leq f(x, y) \leq L_{\max}$
- $[L_{\min}, L_{\max}]$ often mapped to $[0, L-1]$ (for digital image)
- $[0, L-1] \Rightarrow L$ quantization levels
- 0 is called black level, $L-1$ is called white level
- We choose $L = 2^k$, where k is the number of bits required
- 256 levels = 2^8 , hence $k = 8$, called 8-bit image (most common)

- For an image with M rows and N columns
 - $x = 0, 1, 2, \dots, (M - 1)$ where M is the **height** of the image (also called H)
 - $y = 0, 1, 2, \dots, (N - 1)$ where N is the **width** of the image (also called W)
- Numerical array form $[f(x, y)]$
- (i, j) th pixel value $[f(i, j)]$ is the image intensity at point (i, j)
- In Python - `img[x, y]`, `img.shape = (H, W)`

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} a_{0,0} & a_{0,1} & \dots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \dots & a_{1,N-1} \\ \vdots & \vdots & & \vdots \\ a_{M-1,0} & a_{M-1,1} & \dots & a_{M-1,N-1} \end{bmatrix}$$



Coordinate Convention

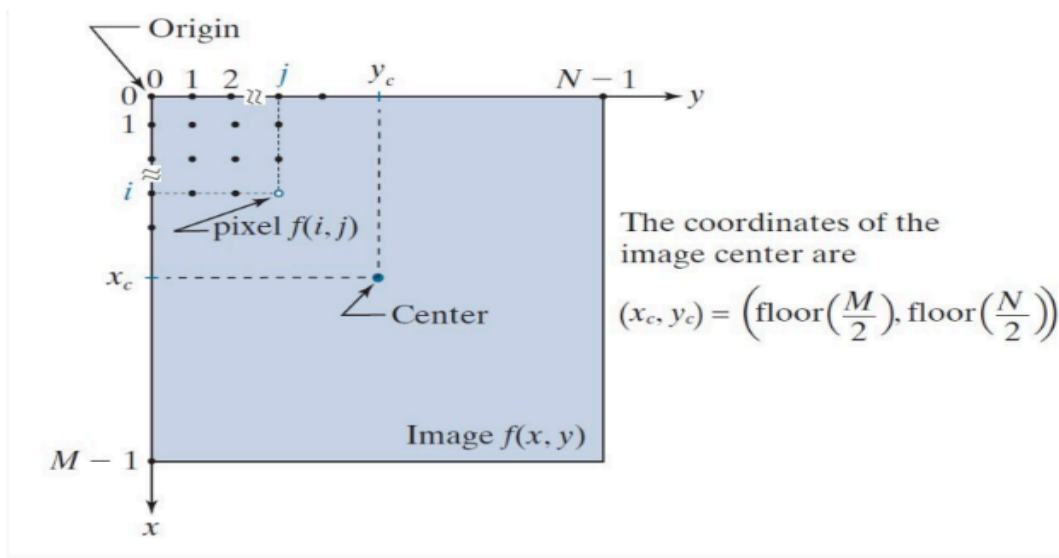


Image Resampling

♦ Downsampling

Downsampling means **reducing the number of pixels in an image**, i.e., lowering its **spatial resolution**.

Basic Idea

- Original image: large size (e.g., 4000×4000)
- After downsampling: smaller size (e.g., 1000×1000)

Fewer pixels $\rightarrow$ less detail

Advantages

- Reduces **image size**
- Faster processing
- Less memory required

Disadvantages

- Loss of **details**
- Can cause **aliasing (distortion)**

- k times downsample $\Rightarrow$ take the k'th sample
- In Python - `inp[y, x, n]`, `inp.shape = (H, W, N)`
- `out[y, x, n]`, `out.shape = (H//k, W//k, N)`
- `for y in range(0, H, k):`
 `for x in range(0, W, k):`
 `out[y//k, x//k, :] = inp[y, k, :]`
- Or, thanks to numpy slicing, `out = inp[::k, ::k, ]`

♦ Upsampling and Interpolation

1. Upsampling

Upsampling means **increasing the image size (resolution)** by adding more pixels.

- Input: low-resolution image
- Output: higher-resolution image

Example:

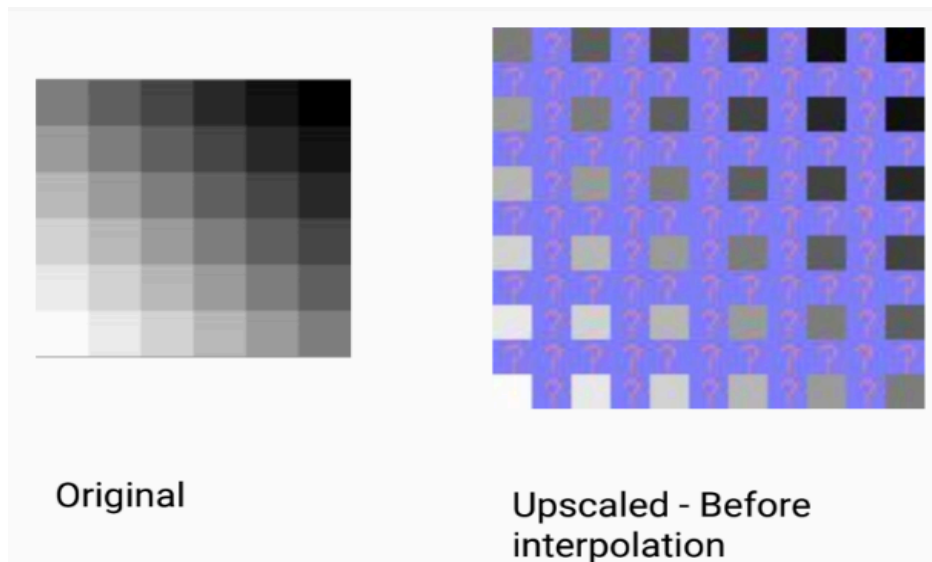
$(100 \times 100) \rightarrow (200 \times 200)$

2. Interpolation

Interpolation is the method used to **estimate values of new pixels**. It fills in the missing pixel values based on neighboring pixels.

Common Interpolation Methods:

1. Nearest Neighbor Interpolation
2. Bilinear Interpolation
3. Bicubic Interpolation

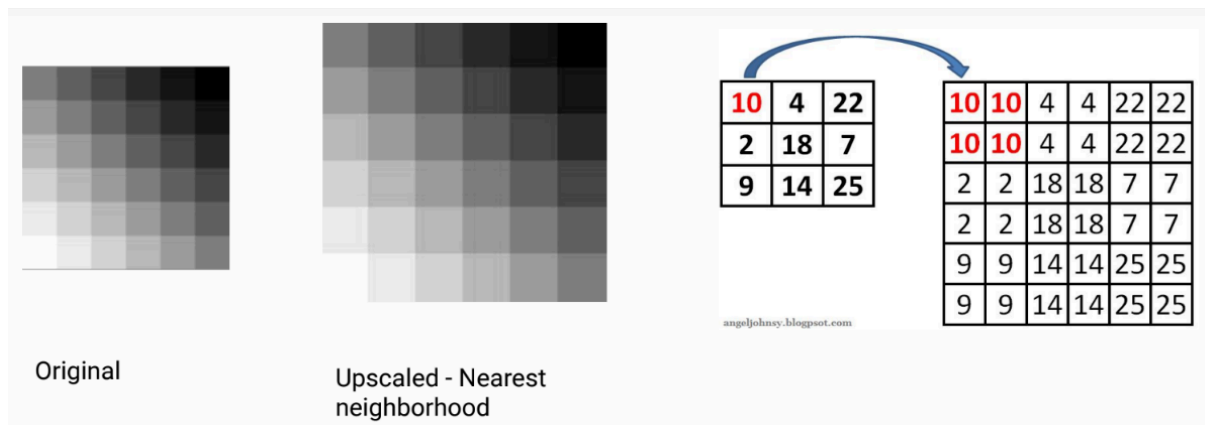


1. Nearest Neighbor Interpolation

- Assign value of the **nearest pixel**

Features:

- Very simple and fast
- Produces **blocky (pixelated) image**

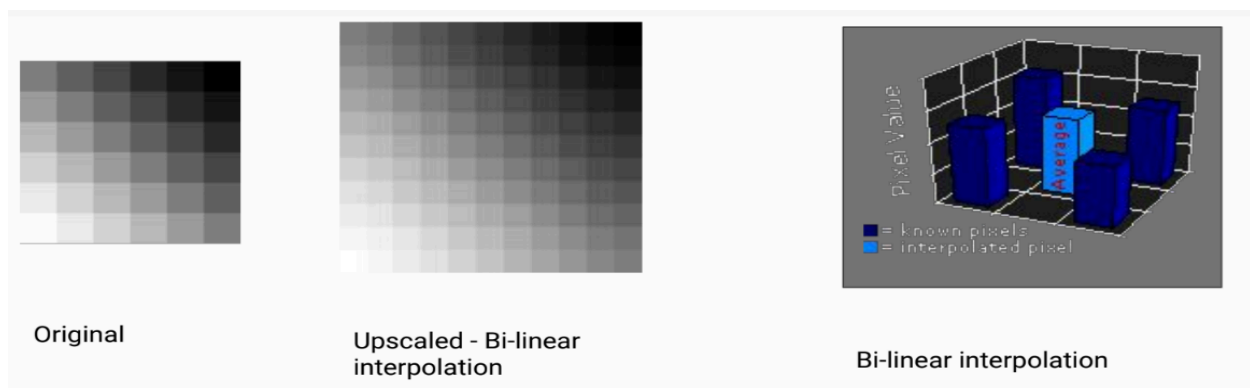


2. Bilinear Interpolation

- Uses **4 nearest neighbors**
- Takes a **weighted average**

Features:

- Smoother than nearest neighbor
- Slight blur



3. Bicubic Interpolation

- Uses **16 neighboring pixels**
- More complex weighting

Features:

- Much smoother and sharper
- More computation

